Statistics Practice for Next week!

Can we conclude that, on average, lymphocytes and tumor cells differ in size? The following are the cell diameters (μm) of 40 lymphocytes and 50 tumor cells obtained from biopsies of tissue from patients with melanoma. Assume your data to be normally distributed.

Lymphocytes							
9.0	9.4	4.7	4.8	8.9	4.9	8.4	5.9
6.3	5.7	5.0	3.5	7.8	10.4	8.0	8.0
8.6	7.0	6.8	7.1	5.7	7.6	6.2	7.1
7.4	8.7	4.9	7.4	6.4	7.1	6.3	8.8
8.8	5.2	7.1	5.3	4.7	8.4	6.4	8.3

Tumor cells									
12.6	14.6	16.2	23.9	23.3	17.1	20.0	21.0	19.1	19.4
16.7	15.9	15.8	16.0	17.9	13.4	19.1	16.6	18.9	18.7
20.0	17.8	13.9	22.1	13.9	18.3	22.8	13.0	17.9	15.2
17.7	15.1	16.9	16.4	22.8	19.4	19.6	18.4	18.2	20.7
16.3	17.7	18.1	24.3	11.2	19.5	18.6	16.4	16.1	21.5

In a study, researchers examined bone strength. They collected 10 cadaveric femurs from subjects in three age groups: young (19-49 years), middle aged (50-69 years), and elderly (70 years or older). One of the outcome measures (W) was the force in Newtons required to fracture the bone. The following table shows the data for the three age groups. Can you conclude that there is a difference among the bone strength among the above age groups? If there is, which group(s) is/are different than the others?

Young (Y)	Middle Aged (MA)	Elderly (E)
193.6	125.4	59.0
137.5	126.5	87.2
122.0	115.9	84.4
145.4	98.8	78.1
117.0	94.3	51.9
105.4	99.9	57.1
99.9	83.3	54.7
74.0	72.8	78.6
74.4	83.5	53.7
112.8		96.0

The following table shows the vital capacity measurements of 60 adult males classified by occupation and age group. Test for differences among occupations, for differences among age groups, and for interaction. Let $\alpha = 0.05$ for all tests; assume your data is normally distributed.

	Occupation			
Age group	**A**	**B**	**C**	**D**
1	4.31	4.68	4.17	5.75
	4.89	6.18	3.77	5.70
	4.05	4.48	5.20	5.53
	4.44	4.23	5.28	5.97
	4.59	5.92	4.44	5.52
2	4.13	3.41	3.89	4.58
	4.61	3.64	3.64	5.21
	3.91	3.32	4.18	5.50
	4.52	3.51	4.48	5.18
	4.43	3.75	4.27	4.15
3	3.79	4.63	5.81	6.89
	4.17	4.59	5.20	6.18
	4.47	4.90	5.34	6.21
	4.35	5.31	5.94	7.56
	3.59	4.81	5.56	6.73

A study of young children examines the effectiveness of an FM sound field when teaching phonics to children. In the study, children in a classroom with no phonological or phonemic awareness training (control) were compared to a class with phonological and phonemic awareness (PPA) and to a class that utilized phonological and awareness training and the FM sound field (PPA/FM). A total of 53 students from three separate preschool classrooms participated in this study. Students were given a measure of phonemic awareness in preschool and then at the end of the first semester of kindergarten. The improvement scores are listed in the following table as measured by the Yopp-Singer Test of Phonemic Segmentation. Test for a significant difference among the three groups. Let $\alpha=0.05$.

Improvement (control)		Improvement (PPA)	Improvement PPA/FM	
0	1	2	1	19
0	1	3	3	20
1	2	15	7	21
4	2	18	9	21
5	3	19	11	22
9	6	20	17	22
9	7	5	17	15
13	8		17	17
18	9		18	17
0	18		18	19
0	20		19	22
-1			19	